

PRANJAY SIDHWANI

+91 9399693258 pranjaysidhwani@gmail.com [LinkedIn](#) [Github](#) [Codeforces](#)

Education

Indian Institute of Technology, Delhi

Bachelors in Technology and Minor in Computer Science

2021 - 2025

8.11/10 CGPA

Scholastic Achievements

- **Minor in Computer Science:** Completed **28 credits** of CS courses, focusing on **Data Structures** and **advanced AI/ML concepts**.
- **National Finalist, American Express Data Science Challenge:** Ranked in top **9** in India out of **4,000+** competing teams.
- **Department Rank 9:** Ranked **9th** out of 95 students in the Textile Technology department based on CGPA till 6th semester.
- **Expert, Codeforces (ID: pranjaysidhwani):** Achieved a **1600+** rating, demonstrating strong skills in competitive programming

Work Experience

Deutsche Bank

July 2025 - Current

Graduate Analyst - Corporate Banking

Pune, India

- Contributed to TF-3, Deutsche Bank's global **trade finance** and corporate lending platform, delivering enhancements and stability gains.
- Developed features and resolved **30+ production bugs**, addressing SonarQube and OSS vulnerabilities to improve code quality.
- Built AI-powered PR review tool saving **1 hour per PR for 100+ developers**, reducing manual effort by **90%**.
- Migrated automation framework from **UFT to Python**, leveraging AI to achieve **90%** faster development and improved scalability.

Internship Experience

Riskspan

Jan 2025 - May 2025

AI Product Intern

New Delhi, India

- Built Smartmapper ETL tool using **BERT**, **fuzzy logic**, and **AWS Bedrock** to automate mapping, reducing manual effort by **70%**.
- Designed mapping pipeline with confidence scoring and dynamic prompts for value resolution and **streamed LLM** outputs.
- Developed **RAG** pipeline with **AWS Bedrock** and **ChromaDB** for citation-backed financial document **Q&A**.
- Optimized Cascaide platform through **LLM caching & embedding refinements**, cutting costs by **35%** and response time by **40%**.

Minfy Technology (Partner AWS)

May 2024 - Aug 2024

AI Product Intern

Wilmington, United States (Remote)

- End-to-End Deployment of **Gen-AI Medical chatbot** for Informia using **Retrieval Augmented Generation** and **AWS** services.
- Custom multi-client chatbots for DTDC, Vedanta, and Dr. Reddy's Labs with **PostgreSQL**, tailored to diverse client needs.
- Developed API using **FastAPI** for chatbot operations and feedback, focusing on **reinforcement learning** through **feedback loops**.
- Enhanced performance by integrating **relevance scores** and **conversation memory**, boosting chatbot effectiveness.

IMT Atlantique

May 2023 - July 2023

Research Intern

Rennes, France (Onsite)

- Enhanced **UAV deployment** by optimizing coverage in disaster management, ensuring more effective monitoring across larger areas.
- Developed strategies for **autonomous drone** coordination, enhancing target coverage, and maintaining robust **multi-hop connectivity**.
- Implemented algorithms for **efficient drone deployment**, dynamically adapting to target densities and geographic challenges.

Projects and Hackathon

[Amex Data Science Challenge](#) - Rank 9/4000 | *Machine Learning*

(May 2024 - Aug 2024)

- Achieved **72.6%** accuracy in predicting cricket match outcomes using **LightGBM** and **feature engineering** on historical match data.
- Combined **Hidden Markov Models** with LightGBM to predict probabilities, merging probabilistic modeling with feature insights.
- Used **SHAP** values to enhance model interpretability, identifying crucial factors like past performance and player statistics.

[TATA 1Mg Project Trainee](#) (Prof. Chetan Arora) | *Computer Vision*

(Jan 2024 - Apr 2024)

- Digitized **handwritten prescriptions** using fine-tuned **Kraken** and **ARUnet** models, achieving **68%** precision and **65%** recall.
- Applied **semantic segmentation** for pixel-level text classification, enabling accurate extraction from complex handwritten documents.
- Outperformed object detection baselines (**Dptext**, **Dbnet**) in complex text scenarios for medical document recognition.

[Correction Agent for ASR Errors in Voice-Enabled Assistants](#) | *Local Search*

(Sep 2023 - Jan 2024)

- Developed an ASR error-correction agent to handle phonetic substitutions and boundary word omissions in voice assistant transcriptions.
- Used **simulated annealing** with **tabu search**, phoneme-based substitutions, and **Whisper NLL** scoring to optimize ASR corrections.
- Improved transcription coherence by minimizing audio-conditioned **negative log-likelihood**, enhancing downstream intent accuracy.

[Eightfold.ai Innov8 Challenge](#) | *Natural Language Processing*

(Oct 2023 - Nov 2023)

- Developed a **Resume Keyword Highlighter** using **Gemma model** API from Hugging Face to extract relevant keywords for job roles.
- Used **cosine similarity** and **TF-IDF** to highlight key resume sections with color-coded highlighting using **PyMuPDF**.
- Deployed the ML model on a **Node.js** web platform for real-time resume analysis, ensuring smooth integration and user interaction.

Technical Skills and Courses

- **Programming Languages:** Python, Java, C++, Matlab, SQL || **Developer Tools:** Postman, VS Code, GitHub, Bruno
- **Developers Frameworks:** Docker, Kubernetes, Langfuse, LiteLLM, Springboot, Maven, Sonarqube, Jfrog, Jenkins, Openshift
- **AWS Cloud Services:** EC2, S3, Lambda, RDS, EBS, VPC, IAM, CloudFront, CloudWatch, SageMaker, SQS, SNS
- **AWS Databases:** RDS, MySQL, PostgreSQL, MongoDB, Redis, Elasticsearch, DynamoDB